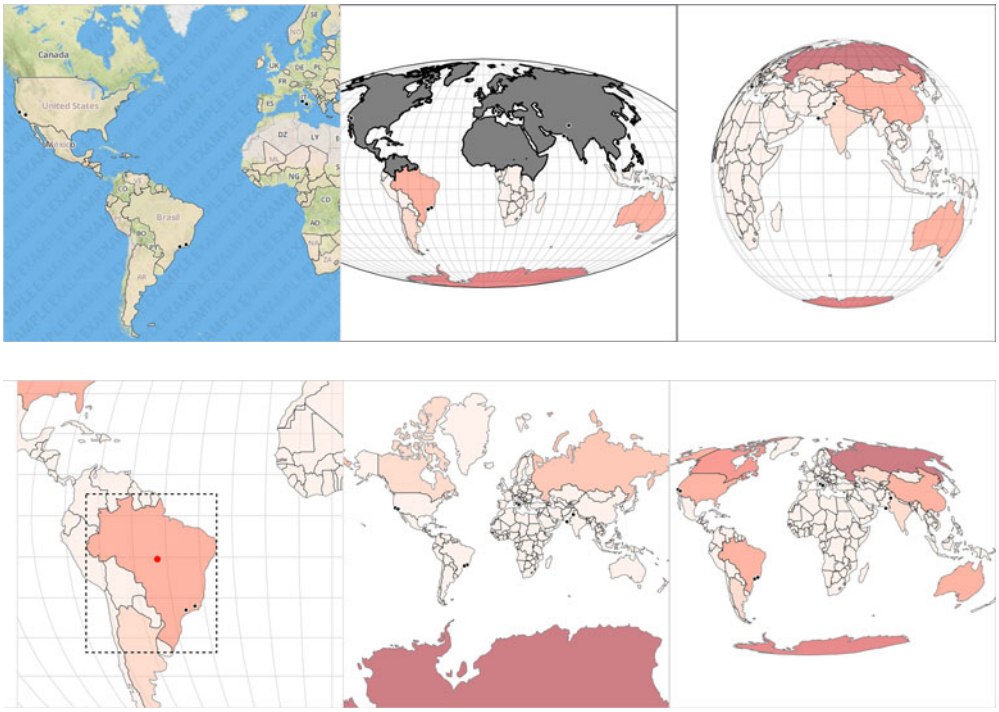


Figure 7.1 Mapping with D3 takes many forms and offers many options, including traditional tile-based maps (section 7.5), cutting-edge TopoJSON operations (section 7.4), globes (section 7.3.1), spatial calculations (section 7.1.4), and data-driven maps (section 7.1) using novel projections (section 7.1.3).

a dedicated library like Google Maps API is because of the added functionality that comes from being in that ecosystem, such as Street View of Google tiles or integrated support for Fusion Tables. But if you're not going to use the ecosystem, then it may be a smarter move to build the map with D3. You won't have to invest in learning a different syntax and abstraction layer, and you'll have the greater flexibility D3 mapping affords.

Because mapmaking and geographic information systems and science (known as GIS and GIScience, respectively) have been in practice for so long, well-developed methods exist for representing this kind of data. D3 has built-in robust functionality to load and display geospatial data. A related library that you'll get to know in this chapter, TopoJSON, provides more functionality for geospatial information visualization.

In this chapter, we'll start by making maps that combine points, lines, and polygons using data from CSV and GeoJSON formatted sources. You'll learn how to style those maps and provide interactive zooming by revisiting `d3.zoom()` and exploring it in more detail. After that, we'll look at the TopoJSON data format and its built-in functionality that uses topology, and why it provides significantly smaller data files. Finally, you'll learn how to make maps using tiles to show terrain and satellite imagery.